

1. SUMMARY

The pelagic trawler, ANTARES left her home port of Carradale on 19 November 1990 to fish in Bute Sound, northeast of the Isle of Arran, with a crew of four. At the same time the Trafalgar-class submarine, HMS TRENCHANT, was operating in the Clyde exercise area. She was acting as a training vessel for officers on the submarine command course and the complement included the Commanding Officer of the course and four student officers. The submarine was taking part in a series of exercises for the final sea phase of the course. The procedure for each exercise series was that conduct of TRENCHANT was passed by her Captain to the Course Commander who in turn gave charge of the submarine to the student to be assessed. The student, designated Duty Captain, then carried out all the functions of command while being observed by the Course Commander.

On 22 November shortly after 0200 hrs TRENCHANT was in Bute Sound, submerged at a depth of 60 metres and steering a course of northwest at about six knots. An exercise series had just been completed and the Duty Captain was preparing to hand over to the next student, who was with him in the control room. The Captain and Course Commander were in the wardroom, discussing the Duty Captain's performance. TRENCHANT was detecting surface vessels by means of her passive sonar.

At 0217 hrs TRENCHANT had a close sonar contact to starboard and turned to port to avoid it. Banging noises were then heard in the submarine and it was assumed by those in the control room that a fishing trawl had been snagged. On regaining periscope depth, two fishing vessels were seen, apparently engaged in normal fishing. TRENCHANT surfaced at about 0300 hrs and a trawl wire was discovered fouled on the submarine casing. Attempts by the submarine to contact the two fishing vessels by radio were unsuccessful. Since everything on the surface appeared normal, TRENCHANT reported the incident to Faslane base and continued with her exercises.

Later in the morning it was reported that ANTARES was missing and a full scale search operation, coordinated by Clyde Coastguard, was mounted. The wreck of ANTARES was located on the sea bed in the early afternoon, the position was where the collision with the trawl had occurred. The wreck was salvaged on 10 December. It was found that the starboard trawl warp had parted and the broken end matched the section of wire found on TRENCHANT's casing.

The bodies of the four crew of ANTARES have been recovered.

The main findings of the investigation are that the accident was caused by a partial breakdown in both the structure and the standards of watchkeeping on board TRENCHANT, following the completion of a command course exercise. Also the stowage position and securing arrangements of the inflatable liferaft on ANTARES were deficient, such that it failed to float free and inflate when the vessel sank.

Further, the delay of over eight hours prior to the mounting of a search operation was due to an incorrect report from TRENCHANT, which stated that although a trawl had been snagged, the fishing vessel involved was safe.

4. NARRATIVE

All times are in GMT.

- 4.1 The crew of ANTARES joined the vessel from their homes around 2300 hrs on Sunday, 18 November 1990, and subsequently sailed from Carradale to trawl the deep trench situated in Bute Sound to the NE of the Isle of Arran. These fishing trips usually lasted from Monday to Thursday with the vessel returning to her base at the weekends. A Scottish Office directive prohibits commercial fishing in the Lower Clyde from 1800 hrs on Friday to 2400 hrs the following Sunday.

There were three other boats in the same general area on 19 November 1990: HEROINE out of Troon, HERCULES III out of Ayr and another called BRIGHTER MORN. BRIGHTER MORN left the area either on the evening of 19 November or the morning of 20 November, while the other three boats remained. There was sporadic VHF communications on Channel 72 between all three vessels throughout that week. HEROINE and HERCULES III remained trawling in the area the whole time but ANTARES may have occasionally visited other areas to look for better fishing and additionally landed her catch in Largs on the evening of 20 November and again on the evening of 21 November.

- 4.2 On 21 November ANTARES left Largs at 1730 hrs and returned to Bute Sound about one hour later. She entered the deep trench towards the southerly end and shot her gear to the south and east of HEROINE and commenced towing NW'ly astern of HEROINE. Once the tow had commenced there was a VHF radio conversation between the Skippers of ANTARES and HEROINE. During this conversation the Skipper of ANTARES stated that it was his intention to tow north and when he turned at the northerly end of the tow he was going below and one of the crew would be taking over. The three fishing vessels continued to fish without further communication.

Just before 2230 hrs the crew member who had taken over the watch from the Skipper made his usual telephone call home to his wife and confirmed to her that they were fishing at what he described as "the back of Arran". He also mentioned that he would be calling the Skipper at 0230 hrs. This is the last known communication with ANTARES.

- 4.3 The three fishing vessels continued to fish normally for the rest of the evening of 21 November and into the early hours of 22 November until some time shortly after 0200 hrs HEROINE, towing NW'ly, passed ANTARES, towing SE'ly, about 2 cables port to port in approximate position 55°40'.3N 5°03'.7W. The time is not certain; it was recalled by the Skipper of HEROINE as "approximately" 0140 hrs. At a time recalled as some 10 minutes after passing ANTARES the watchkeeper in HEROINE looked astern and noted ANTARES turning to starboard to come back on a NW'ly tow. This is the last known sighting of ANTARES. The watchkeeper did not look astern again as he was concentrating on HERCULES III who was ahead of him and towing SE'ly towards him.

A further 10 minutes after this last sighting of ANTARES there was a VHF conversation between the watchkeeper on HEROINE and the Skipper of HERCULES III when the fact that ANTARES could no longer be seen either on radar or visually was discussed. The time of this call was given as 0200 hrs. As there is firm evidence that the accident occurred very close to 0219 hrs, it is considered the times given by the fishermen may be in error, (see Section 8.1). It was assumed by those on board HEROINE and HERCULES III that ANTARES had proceeded south out of the area and no further thought was given to the absence of ANTARES at that time. HEROINE and HERCULES III continued to fish normally.

- 4.4 The weather was excellent with light variable winds and a calm sea: it was a clear night with visibility of about 8 miles. The sea temperature was about 11°C and the air temperature over the water was 6°C. There was no significant tidal stream in the area.
- 4.5 TRENCHANT sailed from the Clyde Submarine Base at Faslane on 12 November to act as a training vessel for Officer students on a command course. This course is referred to as a Perisher course where the students are assessed on their suitability for command. Six such students, together with their Commanding Officer, joined TRENCHANT by boat transfer in Loch Ewe on 14 November. TRENCHANT then proceeded on exercise to a zone between Scotland and the Faeroe Islands.

Prior to each series of exercises, 'conduct' of the submarine was passed by her Captain, to the Course Commander, who is known as 'Teacher', who in turn gave charge of the submarine to the student to be assessed for the duration of his exercise. These exercises required the student to carry out all the functions of command while being observed by 'Teacher'. This is normal procedure for the final sea phase of the command course.

- 4.6 The first phase of exercises was completed on 16 November and TRENCHANT entered the Clyde Exercise Area at 1000 hrs on 17 November to continue with the inshore phase at which time the students onboard TRENCHANT had been reduced to four. This phase of the exercises continued normally until at 1800 hrs on 21 November one of the Perisher students took over as Duty Captain.

He was required to carry out a dummy mine laying exercise during which opposition was to be provided by the surface naval vessel CHARYBDIS. This exercise which was carried out in the Hunterston Channel commenced at 2330 hrs and he was in the control room continuously from that time onwards. This exercise was completed successfully and TRENCHANT proceeded south out of the Hunterston Channel to allow detection by CHARYBDIS so that evasive manoeuvres could be practised.

This phase of the exercise, which involved frequent alterations of course and speed by TRENCHANT continued and at 0100 hrs on 22 November a change of watch took place, but the same student continued as Duty Captain.

- 4.7 The exercise, involving evasive manoeuvres to avoid detection by and to attack CHARYBDIS, continued. At 0131 hrs the submarine went to periscope depth in order to observe CHARYBDIS, prior to launching a simulated attack on that vessel. The Duty Captain took an all round look through the periscope and later remembered noticing only the frigate on the surface before the submarine returned to her submerged depth. At 0158 hrs there was a close sonar contact estimated to be at a range of 384 yards. At this time the submarine was completing a 180° turn to port.

The exercise was declared complete by 'Teacher' just before 0200 hrs. By this time TRENCHANT was steadied on a course of 150°T at about 6 knots at a submerged depth of 60 metres (the keel of TRENCHANT was 60 metres below the surface). Her approximate position at 0200 hrs was 55°39'.5N 5°02'.4W.

- 4.8 When the exercise was declared complete CHARYBDIS steamed in a NW'ly direction out of the area towards the Cock of Arran to engage another submarine, HMS OTTER, which was engaged in a similar exercise to the north. 'Teacher' instructed the Duty Captain to hold in the general area, hand over to his relief and let his relief start things up again from here. 'Teacher' left the control room at this time to discuss the performance of the Duty Captain with the Captain of TRENCHANT in the ward room. The next student to be Duty Captain was then sent to the control room. At 0204 hrs the Duty Captain altered course to starboard and at 0209 hrs TRENCHANT was steadied on a course of 315°T. Speed was still about 6 knots and depth still 60 metres. When TRENCHANT steadied on 315°T there were four apparent contacts on the 2020 Sonar trace - contact MO1 which was the surface naval vessel CHARYBDIS presently steaming away from the area, contact 03 which was classified as a fishing vessel, contact 05 also classified as a fishing vessel and contact 45 which was classified as a coaster but could also have been a fishing vessel.

The contact relevant to the sequence of events which occurred was contact 05 and it was fine on the port bow bearing 304°T. The source of this contact must in fact have been not one but two vessels, namely ANTARES and HEROINE who had just passed each other. It must be remembered that the passive sonar in use gave no firm indication of range, and both craft will have been on similar bearings. The Sonar Controller in the forward sonar room reported contact 05 as being a noisy contact and bearing rate steady. (These sonar reports may be broadcast through loudspeakers in the control room and communications between the sonar room and the control room in this mode are referred to as being 'on the net'. There is conflicting evidence as to whether or not the 'net' was switched on). At 0213 hrs the Sonar Controller reported contact 05 as bearing rate steady but with increased intensity. At 0216 hrs contact 05 was reported as bearing 306°T, moving slowly right and marking 35dB on the noise meter. Between 0216 hrs and 0217 hrs the bearing of contact 05 altered from 306°T to 322°T when it was then reported as getting louder but still marking 35dB on the noise meter and shortly afterwards reported as starting to move fast right. During this period the Duty Captain was in discussion with the student who would be the next Duty Captain. In response to this series of reports and the close contact to starboard the Duty Captain ordered 15° port rudder and later increased this to 30° port rudder. At 0218 hrs TRENCHANT's head was passing

300°T and contact 05 was bearing 355°T. The closest point of approach (CPA) of contact 05 which had already passed was reported by the Operations Officer as 200 yards. At this time, 0218 hrs, the rudder was eased to port 10° and the order to steer 260°T given by the Duty Captain. About 45 seconds later, just before 0219 hrs when contact 05 was bearing 028°T, banging noises on the outside hull of TRENCHANT were heard in the control room. These noises appeared to come from starboard forward.

- 4.9 The Sonar Controller reported almost immediately that a trawl had possibly been snagged and the order to stop engines was given from various sources. The fact that a trawl had been snagged was confirmed shortly afterwards by the noise of the trawl wires dragging down the outside of the submarine. This report was also given by the Sonar Controller. On hearing the first noises the Navigating Officer put the position of TRENCHANT on the chart using the CP inertial navigation system. This position which was considered accurate to 1½ cables was 55°39'.5N 5°03'.3W. The course being steered by TRENCHANT when the collision occurred was estimated as 285/290°T.
- 4.10 The Officer of the Watch then gave the order to stand-by to return to periscope depth and the drill for this was put into operation. This drill included the manning of the 2008 underwater telephone by an Able Seaman. Prior to this the equipment had been unmanned. 'Teacher' and the Captain of TRENCHANT then entered the control room in response to the noises which they had heard themselves and a verbal report from the Navigating Officer. The time was now 45 seconds after collision.
- 4.11 After a brief check 'Teacher' took charge of TRENCHANT from the Duty Captain and proceeded to go slow ahead (5 knots) with 30° of port rudder steadying TRENCHANT on a course of 090°T at 0222 hrs. Her approximate position at this time was 55°39'.5N 5°04'.0W. During this manoeuvre the depth was decreased to 37 metres. 'Teacher' did not want to bring TRENCHANT to periscope depth until he was certain the position of the surface craft made it safe to do so. Checks were then made on the surface contacts and attempts made to regain the previous contact 05 without success. The Able Seaman at his post on the underwater telephone heard a noise which he described as like 'a disturbance of the sea and amidst that noise a sound like a propeller winding up'. This secondary noise lasted only 5 or 6 seconds. There is some suggestion that the command team were standing behind the Able Seaman at this time listening to the same noise. The best possible estimate of time for this event is between 0221 hrs just before TRENCHANT steadied on 090°T and 0226 hrs when she passed over what is now known to be the wreck position.
- 4.12 'Teacher', after taking further evasive action to distance TRENCHANT from any close surface contact, eventually brought the submarine to periscope depth at 0239 hrs. After viewing the surface situation through the periscope he reported two fishing vessels in close proximity to each other but no close contacts to the submarine itself. He ordered Sonar Standard (this means they could relax somewhat since the surface situation was held on visual) and handed conduct of the submarine to the Captain of TRENCHANT to complete the surfacing operation.

Whilst at periscope depth the radio mast was raised and contact made with CHARYBDIS to ask if anything unusual had been heard from fishing boats. The reply was negative.

- 4.13 The Captain brought TRENCHANT to the surface at 0252 hrs in position 55°39'.5N 5°01'.0W. An inspection at this time revealed a wire with a length of chain attached fouled around the 'Paris' dome of the submarine. (This is a small dome forward of the fin which houses an active Sonar). This wire started on the starboard side of the dome where the attached chain was embedded in the glass reinforced plastic, ran around the forward end of the dome and down into the water on the port side leading aft with considerable weight on it. This wire was finally removed by a casing party when the weight came off it, but to achieve this the way had to be taken completely off TRENCHANT using astern power before the wire could be cut with bolt cutters when it fell to the sea bed. The short length of wire and the attached chain remaining was retained as evidence.

Once on the surface 'Teacher' again made radio contact with CHARYBDIS to ascertain if they had seen or heard anything unusual and the response was negative. He then called, on VHF radio, the two fishing vessels he had seen earlier through the periscope. Channel 16 and several other channels were used for these calls, which were made without formal prefixes. No response was received from the two fishing vessels which were HEROINE and HERCULES III. At about 0315 hrs 'Teacher' contacted the Operations Room at the Faslane submarine base and reported the incident.

TRENCHANT remained on the surface while the overall situation was assessed by the Captain. While on the surface one of the two fishing vessels passed close by without any contact being made. The Captain finally decided to advise Faslane that TRENCHANT should continue with the exercise. He based this decision on the apparent normality of the surface situation, the fact that no alarm had been raised and everything seemed normal with the two fishing boats in the area and the minimal damage only to his own submarine. This advice was passed to Operations at Faslane and TRENCHANT submerged at 0500 hrs to continue with the exercise.

- 4.14 At 0900 hrs CHARYBDIS completed her exercise with OTTER and steamed south down the Firth of Clyde out of the area. TRENCHANT completed the final phase of the exercise programme and returned to the Faslane base arriving there at approximately 1200 hrs. It was only at this stage and less than one hour before berthing that the command on board TRENCHANT started receiving indications that there might be a fishing vessel missing.
- 4.15 Whilst at Faslane TRENCHANT was put in dry-dock and in addition to the damage to the 'Paris' dome where the wire had embedded itself the following damage was also noted:

- a) The starboard log probe was sheared off.
- b) The pinger and its protective cowling sited centrally on the keel plate towards the forward end of the submarine were damaged with the holding down bolts at the forward end ripped away and the base plate badly bent.
- c) The leading and trailing edges of the starboard forward fin were damaged.
- d) Minor damage to the starboard side in way of the starboard forward fin.
- e) There were rubber panels missing from the starboard side of the hull not thought to be connected with the incident.

9. RECONSTRUCTION OF EVENTS

- 9.1 The normal behaviour of pelagic trawlers fishing in Bute Sound is to tow up and down the trench only turning at the northerly and southerly limits of the tow. When turning, the procedure is to first increase speed to normal full ahead to raise the net from the bottom. Once this is done, an alteration of course of 90° is made quickly using hard over rudder and then steadying on this course until the weight comes back on the trawl gear. When this occurs, a further alteration of 45° is made and then steadying on this course until once again the weight is on the trawl gear, and then complete the 180° turn with a further 45° alteration. The first part of this turn from increasing speed to steadying on a course of 90° from the original takes about 4 minutes and the complete turn from start to finish about 15 minutes. When carrying out this procedure by turning to starboard, both warps would lead out taut over the starboard quarter. The warp on the outer part of the turn, in this case the port one, takes more strain and the trawl door attached to it 'flies' higher in the water than the starboard one so that the port warp would be higher in the water than the starboard warp. Additionally, during the turn the net closes bringing both warps close together and a constant watch has to be kept on them to avoid them crossing each other. These trawlers are very vulnerable when turning and the Skipper is always on the bridge at this time.
- 9.2 From the evidence of the wreck of ANTARES, that is with the engines at normal full ahead, rudder hard to starboard, starboard wheelhouse window opened and the searchlight switched on, it is probable that she was executing a normal turn at the southerly end of her tow. The open window and searchlight would be necessary to see that the warps did not cross each other during the turn.
- 9.3 ANTARES was held on a near steady bearing by TRENCHANT from 0209 hrs until 0216 hrs when something happened on board ANTARES to make the bearing change and start to draw right. The reported increase in noise level to 35dB at this time is possibly indicative of the time that ANTARES increased to normal full speed to raise the net from the bottom. This would have caused the bearing to start to draw right.

This implies that ANTARES was in the first part of her turn when the collision occurred and this is further corroborated by the previous steady bearing of contact 05 on board TRENCHANT. The very fact that the bearing remained steady indicates that the course and speed of ANTARES during that period did not alter and this would not have been the case if ANTARES had been engaged in her turn at any time prior to 0216 hrs.

- 9.4 Consideration of the evidence indicates a course for ANTARES during the last few minutes before she began her turn of about 102°T. The natural course for a SE'ly tow down the trench would be more in the region of 130°T. It is probable that once the net of ANTARES was clear of the net of HEROINE, (after they passed in close proximity), ANTARES was angled over more to the east of the trench so that when it was time to execute the planned turn they would be lined up on the track which had previously been selected for the next

NW tow. It is probable therefore that ANTARES altered course and steadied on a course of 102°T around 0209 hrs. One effect of this will have been to tend to keep ANTARES and HEROINE on about the same relative bearing from TRENCHANT, thus helping to explain the failure to distinguish them as two separate contacts.

9.5 The following is the most probable sequence of events:

ANTARES turned at the northerly end of the trench sometime around 2230 hrs on 21 November and after this turn the watchkeeper took over from the Skipper with the intention of calling him some four hours later for the turn at the southerly end of the tow.

ANTARES then towed SE on an approximate course of 130° until at about 0205 hrs she passed HEROINE (towing NW) port to port, at a distance of 2 cables. Once clear of the nets of HEROINE, ANTARES altered course to 102°T around 0209 hrs to bring the vessel further east in the trench, prior to turning to commence her NW tow.

TRENCHANT meanwhile, having just completed an exercise with CHARYBDIS, altered course from 150°T at 0204 hrs and steadied on 315°T at 0209 hrs. At 0210 hrs the approximate position of TRENCHANT was 55°38'.9N 5°02'.4W. TRENCHANT continued on this course at a speed of about 6 knots until at 0217 hrs a close sonar contact to starboard with ANTARES occurred, when TRENCHANT altered course to port to put more distance between herself and this close contact.

At about this time ANTARES who had gone to full ahead about 0216 hrs commenced to turn to starboard as the first part of a 180° turn back into the NW.

At 0219 hrs TRENCHANT collided with the warps of the ANTARES fishing gear which, at that stage, would be leading broad on the starboard quarter so that ANTARES was almost beam on to them, and the resulting transverse pull caused ANTARES to capsize to starboard and sink in position 55°39'.5N 5°03'.3W. (See Figure 4)

- 9.6 The MoD have also carried out their own investigation into the accident. Their reconstruction of events in respect of the movement of ANTARES and other fishing vessels in the area differs appreciably from that of the MAIB Inspectors given above. The MoD reconstruction is that ANTARES had completed her turn and was on her NW'ly trawl track when the accident occurred. However, both investigations agreed that TRENCHANT altered course to port and collided with the trawl gear of ANTARES at 0219 hrs.

These two reconstructions were both considered by the Sheriff Principal during the Fatal Accident Inquiry held in September 1991 and he regretted that both teams of investigators were unable to agree a joint reconstruction. He did add however it was of very limited significance to the actual outcome of what happened at 0219 hrs, and this point is fully accepted by MAIB.

11. PREVIOUS INCIDENTS

- 11.1 There have been a series of incidents involving submarines and trawlers over the last few years and prior to this particular incident the recorded number involving allied submarines since 1980 was 15. One of these incidents resulted in the foundering of the trawler involved, namely SHERALGA in 1982.

These incidents have led to close co-operation between the Ministry of Defence (Navy), the Surveyor General's Organisation and the Marine Accident Investigation Branch of the Department of Transport in an effort to resolve the problem. This co-operation has led to some useful initiatives.

- 11.2 One such initiative was the feasibility study by the ARE (Admiralty Research Establishment) into the possibility of submarines receiving signals from echo sounders and fish finders of surface craft to assist in the evaluation of such contacts. This resulted from a recommendation in an earlier report. The study concluded that the recommendation could not be accepted because present sonar capacity could not receive all echo sounder frequencies.
- 11.3 A further initiative resulting from the co-operation between the Departments has resulted in tests of a pinger device to assist submarines to detect the presence of trawl nets. It is understood that these tests are well advanced and the outcome of them is awaited. If these tests are successful this would be a useful device in particular for pelagic nets which do not scrape along the bottom and consequently do not make much noise.

12. RESPONSIBILITY

12.1 ANTARES was going about her legal business of commercial fishing in the area and despite the fact that it was a submarine exercise area, there is no possible way that she could have known of the presence of TRENCHANT underwater, and no blame can therefore be attached to ANTARES.

12.2 Royal Navy standing orders, in force at the time, required submarines to "remain clear of fishing vessels and never approach (them) within 2,000 yards"; additionally the standing orders set by the Captain of TRENCHANT specified, that he (or whoever had conduct of the submarine at the time) was to be called if the submarine 'found itself' within 4,000 yards of a surface contact. It was therefore the duty of the command team on board TRENCHANT to ensure that she kept clear of surface contacts and where applicable their attached nets. The fact that she failed to do this is due in part to the exercise which had just been completed. The exercise itself meant frequent alterations of course and speed making a proper evaluation of the surface situation difficult so that at 0209 hrs when TRENCHANT steadied on 315°T, the command team had no clear idea of what the surface contacts were doing.

At this time the bearing of contact 05 steadied at about 304°T fine on the port bow. It does not appear to have been appreciated by the command team that this contact was in fact two vessels, one of which was going away (HEROINE) and the other was coming towards them (ANTARES). An incorrect report at this time (0209 hrs) by the Operations Officer regarding contact 05, to the effect that the bearing was steady and the range opening may have lulled them into a false sense of security so that no serious attempt was made to analyse contact 05 from 0209 hrs until 0216 hrs when it started to move right with increased noise intensity.

12.3 With a steady bearing, no proper calculation of range can be made and in these circumstances the submarine must alter course or speed or both to make the bearing change; range can then be worked out. This is called a ranging manoeuvre and it is understood that such a manoeuvre is required by standing orders when a steady bearing is encountered. No ranging manoeuvre was carried out and it is ironic that the very act of carrying out a ranging manoeuvre in good time would in itself have averted this collision.

12.4 When the exercise was declared complete there was a general lowering of awareness among the watchkeepers in the control room with a certain amount of movement for cups of tea etc. Additionally, the withdrawal of CHARYBDIS could have been a subterfuge and this vessel might have returned to the attack. In these circumstances it is possible that any attention which was being paid to the surface contacts was in fact concentrated on CHARYBDIS. Between 0209 hrs and 0216 hrs there is one report from the Operations Officer on the range, course and speed of CHARYBDIS when she was still going away. There was no attempt whatsoever at serious analysis of any of the other surface contacts.

- 12.5 Just before 0200 hrs, 'Teacher' left the control room with the Captain and went to the wardroom to discuss the performance of the Duty Captain during the previous exercise. When 'Teacher', who had conduct of the submarine at this time, left the control room, the surface situation had still not been evaluated properly and he instructed the Duty Captain to hold in the general area and hand over to his relief. The next Duty Captain was waiting in the wardroom to be called and when 'Teacher' entered the wardroom he sent this officer to the control room to allow the Captain and himself some privacy. When the new Duty Captain entered the control room, he engaged in conversation with the Duty Captain he was to relieve; it is probable that the Duty Captain was not paying complete attention to what was happening around him with the result that, even though a contact was held on a steady bearing fine on the port bow no sense of urgency was transmitted from him to the watch in the control room.
- 12.6 A contributory factor to the lack of any sense of urgency regarding contact 05 was that no trawl noise was reported. This may have lulled the command team into a false sense of security since TRENCHANT was operating at a safe depth of 60 metres. A safe depth is a depth at which it is impossible for the submarine to come into contact with any surface craft, however deep drafted, so that the only possible hazard would have been a net. But the deep trench is fished regularly by the Clyde fishermen and the command team on TRENCHANT should have been aware of this, so that the possibility of encountering a net in these waters ought never to be discounted. Although no trawl noise was reported, the contact was classified as a fishing vessel and in these circumstances the assumption that the boat was fishing should have been made. This was not done, as evidenced by the lack of any alarm in the control room even after the close contact with 05 was experienced.
- 12.7 When the bearing of contact 05 started to move fast right the Duty Captain altered course to port away from it. This alteration of course was ordered without sufficient information regarding contact 05 being available, but it was correctly made in accordance with standing instructions for all submarines, which require a turn away from any close surface contact. It was blind and instinctive but without panic; but in the final analysis it actually caused the collision with the warps since, if no alteration of course had been made by TRENCHANT, it is probable that she would have passed close astern of ANTARES and underneath the warps which at this point would have been at a shallower depth.
- 12.8 There is conflicting evidence regarding the number of hours worked by the Duty Captain for the period in question. According to him he had two hours rest before he came on duty at 1800 hrs on 21 November and this short rest period had been preceded by a considerable period as Duty Captain. According to Navy records he went off duty at 0700 hrs on 21 November and did not take over as Duty Captain until 1930 hrs that day implying that he had a considerable rest period of some 12½ hours. In the circumstances it is difficult to form an opinion as to whether the Duty Captain may or may not have been fatigued; the possibility that he was should not be discounted, but he did not at the time feel impaired from discharging his responsibilities.

PART V CONCLUSIONS

13. FINDINGS

The Inquiry carried out by the Inspectors has covered great detail. It is inevitable in an accident such as this when there are no survivors from the fishing vessel, no direct eye-witnesses on the surface to the accident and the submarine was submerged at the time, that a certain amount of supposition must be taken into account.

During their Inquiry, the Inspectors received the full co-operation of the Ministry of Defence and the Royal Navy, who carried out their own inquiry into the accident. Even so both inquiries came to different conclusions as to the exact circumstances of the collision and the difference in the two reconstructions of the course ANTARES was steering at the time and how TRENCHANT came into contact with the trawl gear is fundamental. At the same time however it was of very limited significance to the cause of events.

However, the suppositions made by the Inspectors to reach their reconstruction are based on logical reasoning and I support their conclusions.

Furthermore, I support the findings of the Inspectors who carried out the Inspectors' Inquiry and these follow:

- 13.1 On 22 November MFV ANTARES was engaged in trawling in the deep trench situated in Bute Sound, to the northeast of the Isle of Arran. Just before 0219 hrs ANTARES was turning to starboard at the end of her southeasterly tow when HMS TRENCHANT collided with the warps of her trawl gear. ANTARES was lying beam on to her gear at this time and the resulting transverse pull caused her to capsize to starboard and subsequently founder, with the loss of all hands, in position 55°39'.5N 5°3'.3W.
- 13.2 The sole cause of the collision was a partial breakdown in the watchkeeping structure and standards on board TRENCHANT.
- 13.3 ANTARES was a seaworthy vessel and her stability was more than adequate for a vessel of her class. However, her life-saving appliances were deficient in that the stowage position and securing arrangements of the inflatable liferaft prevented it from floating free and inflating as the vessel sank. This could have been a contributory cause to the loss of life.
- 13.4 The partial breakdown of watchkeeping procedures on board TRENCHANT was caused mainly by the successful completion of a Perisher exercise. There was some movement among watchkeepers and the general level of alertness was lowered. The command team on board TRENCHANT had no clear appreciation of the surface contacts held on sonar during the period between the completion of the exercise and the collision.

- 13.5 The command team on board TRENCHANT did not fully appreciate that there were two vessels within the sonar contact relevant to this collision going in opposite directions.
- 13.6 No ranging manoeuvre was carried out with respect to the sonar contact on a steady bearing fine on the port bow. This would have established the range of that contact.
- 13.7 The execution of a ranging manoeuvre in adequate time would in itself have averted collision.
- 13.8 More attention was paid by the command team on board TRENCHANT to the sonar contact emanating from the surface naval vessel CHARYBDIS than from the other surface contacts.
- 13.9 The concentration of the Duty Captain with respect to the overall situation was impaired due to his conversation with the next Duty Captain in the minutes before collision.
- 13.10 The Duty Captain may have been fatigued but this in itself did not contribute to the casualty.
- 13.11 The absence of any report regarding trawl noise from the ANTARES sonar contact lulled the command team on board TRENCHANT into a false sense of security and the incorrect assumption that ANTARES was not engaged in trawling.
- 13.12 The alteration of course to port by TRENCHANT in response to a close sonar contact to starboard was made on the basis of insufficient information, but it was the correct action to take in accordance with standing procedures for submarines.
- 13.13 No proper assessment of what might have occurred on the surface was carried out by the TRENCHANT command subsequent to the collision.
- 13.14 The attempts which were made to establish contact with the fishing vessels in the area after TRENCHANT surfaced were not adequate. Standing orders for submarines required communications to be established (by whatever means) after a snagging incident.
- 13.15 The report to the Faslane Operations Room regarding the incident, which indicated that the fishing vessel involved was safe, was made on the basis of insufficient information.
- 13.16 The initial incorrect reports from TRENCHANT led to an approximate 8½ hour delay in a search and rescue operation being mounted. This may have contributed to the loss of life.

- 13.17 TRENCHANT left the scene of the collision and resumed exercise and this decision was based on insufficient information and a lack of appreciation of the reality of the situation.
- 13.18 Clyde MRCC bear no responsibility for the delay in mounting a search and rescue operation which, when finally mounted, was thorough and well co-ordinated.
- 13.19 A listening watch was not being kept on board HEROINE and HERCULES III on Channel 16 VHF. There is no mandatory requirement for such a listening watch, but it is recommended by Admiralty Notices to Mariners. The failure to follow this recommendation may have contributed to the loss of life.